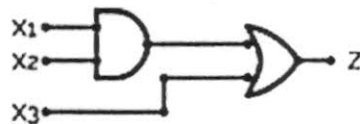


12. What is the key difference between RAM and ROM (Read-Only Memory)?
 A) RAM is writable, while ROM is read-only. B) RAM is used for sequential circuits, while ROM is used for combinational circuits.
 C) RAM is volatile, while ROM is non-volatile. D) Both A and C
13. What is the primary difference between a Moore machine and a Mealy machine?
 A) The number of states B) The type of logic gates used
 C) The output behavior D) The clock frequency
14. In a Moore machine, when are outputs generated or changed?
 A) On state transitions B) On clock edges
 C) On input changes D) On clock edges and input changes
15. What is a common application of a sequence detector in digital design?
 A) Multiplication of binary numbers B) Addition of binary numbers
 C) Detecting specific patterns or sequences of bits in a data stream D) Storing specific sequences in memory
16. In an FSM design, what does a state transition diagram represent?
 A) The order in which inputs are processed B) The behavior of combinational logic
 C) The transitions between states and conditions triggering those transitions D) The clock frequency
17. What does FPGA stand for?
 A) Field-Programming Graphics Adapter B) Field-Programmable Gate Array
 C) Finite-Programmable Gate Arise D) Flexible Program Generation Algorithm
18. What is the primary advantage of using an FPGA over an ASIC (Application-Specific Integrated Circuit)?
 A) Lower cost B) Faster processing speed
 C) Flexibility and re-programmability D) Both are same
19. In FPGA design, what is synthesis?
 A) The process of generating a netlist from a high-level description B) The process of physically manufacturing the FPGA chip
 C) The process of configuring the FPGA with a bitstream file D) The process of testing the FPGA design
20. Which of the following is NOT a step in the FPGA design flow?
 A) Synthesis B) Place and Route
 C) Configuration D) Testing

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I

	Marks	BT*	CO*	PO*
1. a) Explain any Four operators available in Verilog. Give two examples for each.	8	L*2	1	1
b) Design a 2-4 decoder (active low output) using logic gates. Write Verilog code using Dataflow and Behavioral Style. Also, write the test bench.	8	L3	2	1
2. a) Consider a single Verilog code having - assign statements, always block, and initial block (in a single module). Explain the syntax and the flow of the code.	8	L2	1	1
b) Function $F(A, B, C, D) = \sum m(0, 3, 4, 7, 8, 9, 13, 14)$. Write Verilog code for function F using 8:1 MUX with the help of structural style.	8	L3	2	1
3. a) Using the Fault table method, find the minimum number of test vectors required to verify the circuit shown in Fig. 3(a)				
 Fig. 3(a)				
b) For the test vectors generated in Q.3(a), write Test bench using: (i) Normal method- where the test vectors are generated directly, (ii) Using FOR Loop.	8	L3	2	1

Unit – II

- | | | | | | | |
|----|----|---|---|----|---|---|
| 4. | a) | Write Verilog code for a JK flip flop using IF-ELSE and CASE construct. | 8 | L3 | 3 | 1 |
| | b) | Design a sequential block that receives 4-bit data and loads the data serially when ser='1', par='0'. When par='1', ser='0', it loads the data parallelly. Also, when both par=ser=0, it loads zeros. Write Verilog code and explain the working. | 8 | L3 | 3 | 1 |
| 5. | a) | <p>You are given a 4x2 RAM module with the following initial state of the memory array:
[00,01,10,11]
Perform the following sequence of operations:
Write operation with address = 00, data_in = 10, and write_enable = 1
Write operation with address = 11, data_in = 01, and write_enable = 1
Read operation with address = 01, and read_enable = 1
Based on these operations:
Explain the working of this 4x2 RAM module, detailing how each operation modifies the state of the memory array or retrieves data from it.
Write the behavioral Verilog code that would implement this 4x2 RAM module, taking into account the given sequence of operations.</p> | 8 | L3 | 3 | 1 |
| | b) | Write Behavioral Verilog code for a 4-bit up/down counter. When up='1', the down='0' counter works as a UP counter. When up='0', down='1' counter works as a down counter. When reset=1, the counter output is set to Zero. Also, reset should be equal '0' for the counter to work. When the count output to reset when it is equal to "1100" during up-counting. | 8 | L3 | 3 | 1 |
| 6. | a) | Write Verilog code to implement a D flip-flop using behavioral modeling. Include a testbench to simulate its operation. Provide the relevant equations for D flip-flop behavior. | 8 | L3 | 3 | 1 |
| | b) | Create a Verilog module for a clock divider that takes an input clock signal and generates an output clock signal that runs at 1/4th the frequency of the input clock. Explain the working of the Verilog module. | 8 | L3 | 3 | 1 |

Unit – III

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|----|----|---|----|----|---|---|
| 7. | a) | Design a Moore-based binary counter that counts from 0 to 3 (in binary, from "00" to "11"). The output should be '1' only when the counter reaches a count of 3, and '0' otherwise. The counter should reset to 0 after reaching 3 and continue counting. Draw the state transition diagram for the binary counter. | 4 | L2 | 4 | 1 |
| | b) | Write the Verilog code to implement the Moore-based binary counter described in Question 7(a). | 12 | L3 | 4 | 1 |
| 8. | a) | Design a Mealy-based sequence detector that detects the pattern "101" in the input string of binary values. The output should be '1' only when the pattern is detected and '0' otherwise. The sequence can overlap. Draw the state transition diagram for the sequence detector. | 4 | L2 | 4 | 1 |
| | b) | Write the Verilog code to implement the Mealy-based sequence detector described in Question 8(a). | 12 | L3 | 4 | 1 |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
